

|                |                                                                                                                                                       |    |     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|
| <b>2 (a)</b>   | speed = distance/time and velocity = displacement/time                                                                                                | B1 |     |
|                | speed is a scalar as distance has no direction <b>and</b><br>velocity is a vector as displacement has direction                                       | B1 | [2] |
| <b>(b) (i)</b> | <u>constant</u> acceleration or linear/uniform increase in velocity until 1.1 s                                                                       | B1 |     |
|                | rebounds or bounces or changes direction                                                                                                              | B1 |     |
|                | decelerates to zero velocity at the same acceleration as initial value                                                                                | B1 | [3] |
| <b>(ii)</b>    | $a = (v - u)/t$ or use of gradient implied                                                                                                            | C1 |     |
|                | $= (8.8 + 8.8)/1.8$ or appropriate values from line or $= (8.6 + 8.6)/1.8$                                                                            | B1 |     |
|                | $= 9.8 \text{ (9.78) ms}^{-2}$ or $= 9.6 \text{ ms}^{-2}$                                                                                             | A1 | [3] |
| <b>(iii)</b>   | 1. distance = first area above graph + second area below graph                                                                                        | C1 |     |
|                | $= (1.1 \times 10.8)/2 + (0.9 \times 8.8)/2 (= 5.94 + 3.96)$                                                                                          | C1 |     |
|                | $= 9.9 \text{ m}$                                                                                                                                     | A1 | [3] |
|                | 2. displacement = first area above graph – second area below graph                                                                                    | C1 |     |
|                | $= (1.1 \times 10.8)/2 - (0.9 \times 8.8)/2$                                                                                                          |    |     |
|                | $= 2.0 \text{ (1.98) m}$                                                                                                                              | A1 | [2] |
| <b>(iv)</b>    | correct shape with straight lines and all lines above the time axis or all below                                                                      | M1 |     |
|                | correct times for zero speeds (0.0, 1.15 s, 2.1 s) and peak speeds<br>( $10.8 \text{ ms}^{-1}$ at 1.1 s and $8.8 \text{ ms}^{-1}$ at 1.2 s and 3.0 s) | A1 | [2] |